Recent advances in Predictive Learning Analytics: A decade systematic review (2012–2022)

Automated LLM-assisted Review Pipeline

Abstract

This systematic review synthesises research on Predictive Learning Analytics (PLA) in higher education published between 2012 and 2022, drawing on 165 included studies identified from an initial pool of 5,709 records after deduplication, ranking, and screening. The reviewed literature spans eight thematic areas: predictive modelling methods, student performance prediction, retention and dropout, engagement and learning behaviour, data sources and feature engineering, intervention and support, implementation at scale, and interpretability and explainability. A cross-study pattern reveals the dominance of supervised machine learning techniques—particularly Random Forest, Decision Trees, and Logistic Regression—applied primarily to institutional and learning management system data for forecasting student outcomes. While many models achieve high accuracy (often exceeding 80

1 Introduction

1.1 Background

The increasing digitisation of higher education has generated vast amounts of learner data, creating opportunities to forecast student outcomes and intervene proactively. Predictive Learning Analytics (PLA) lies at the intersection of Learning Analytics, Educational Data Mining, and Machine Learning, aiming to identify at-risk students, estimate performance, and model engagement, satisfaction, or enrollment behaviour. Over the past decade, the field has matured from proof-of-concept demonstrations toward more robust, institutionally embedded systems. Concurrent advances in algorithmic complexity—especially the rise of deep learning and ensemble methods—have expanded the predictive toolkit available to researchers and practitioners. Despite this progress, the rapid evolution of methods, evaluation practices, and application domains calls for a systematic synthesis that captures the state of the art and identifies persistent challenges.

1.2 Motivation and gap

Although several reviews have examined specific facets of PLA—such as dropout prediction, performance forecasting, or feature engineering—a comprehensive, decade-spanning overview that simultaneously tracks research trends, intervention contributions, and methodological choices remains lacking. Prior surveys often narrow their scope to a single outcome (e.g., course grades) or a single technique (e.g., decision trees), leaving cross-domain patterns and emerging themes underexplored. Moreover, the field has witnessed a notable shift from traditional machine learning to deep learning architectures, yet the extent to which this shift translates into practical educational impact is not well understood. The absence of an up-to-date, evidence-balanced synthesis hampers the ability of researchers and decision-makers to locate existing evidence, identify mature areas, and pinpoint underexamined gaps.

1.3 Review objective

This study aims to conduct a systematic review of the most recent research body on Predictive Analytics in Higher Education, covering publications from 2012 to 2022. By analysing studies indexed across eight major bibliographic databases, the review seeks to characterise the evolution of PLA research over a critical decade of growth and transformation. The objective is to provide a structured picture of the field's thematic landscape, methodological toolkit, and the ways in which predictive models have been used to support student learning processes.

1.4 Research questions

The review is guided by three research questions:

- **RQ1:** What are the research trends of the PLA field in the last years?
- **RQ2:** How does PLA contribute to supporting students' learning process?
- **RQ3:** What are the ML methods used in conducting PLA?

1.5 Contributions

This review contributes a consolidated evidence base that maps a decade of predictive learning analytics research. Specifically, it (i) delineates temporal and thematic trends in study focuses—from early dropout prediction to broader engagement and performance modelling; (ii) synthesises how PLA interventions have been operationalised to support students, revealing both direct and indirect influence pathways; and (iii) catalogues the machine learning and deep learning methods employed, along with their reported performance characteristics. These contributions are intended to serve as a reference point for researchers planning future work, practitioners seeking evidence-informed tools, and institutional leaders evaluating the readiness of PLA for broader adoption.

2 Methods

2.1 Review design and objective

This systematic review aimed to synthesize recent advances in Predictive Learning Analytics (PLA) within higher education, covering publications from 2012 to 2022. The review addressed three research questions: (RQ1) trends in PLA research over the past decade, (RQ2) how PLA supports student learning processes, and (RQ3) the machine learning methods employed in PLA studies. The review followed a structured protocol for article retrieval, ranking, and evidence extraction.

2.2 Search sources and keyword strategy

Searches were conducted across eight digital libraries and databases: Scopus, Web of Science, Google Scholar, OpenAlex, Crossref, Semantic Scholar, Europe PMC, and PubMed. A broad set of keywords was used, including "Predictive Learning Analytics," "Learning Analytics," "Educational Data Mining," "Machine Learning," "Deep Learning," "Higher Education," "Student Performance," "At risk students," "Engagement," "Satisfaction," "Enrollment," "Predictive modelling," "Artificial Intelligence," "predictive analytics," "student success," "dropout," "at risk," "Predict," "Forecast," "EDM," "Artificial Intelligence in Education," "AIED," "Data Mining," "University," "College," "Learning outcomes," "Learning results," "Student outcomes," "Student results," "grade," "performance,"

"retention," "fail," "motivation," "predictive model," "student outcome," "prediction," "algorithm," and "performance metric". Searches were restricted to English-language, peer-reviewed publications from January 2012 to December 2022.

2.3 Eligibility criteria

Studies were included if they were written in English, peer-reviewed, conducted in a higher education context, employed at least one machine learning or deep learning method, clearly stated predicted learning outcomes, described data collection and preparation, and reported proper validation with clearly defined evaluation measures. Exclusion criteria comprised review, survey, meta-analysis, and commentary articles.

2.4 Article retrieval, cleaning, and ranking

The initial database search yielded 5,709 records (`records_downloaded`). After deduplication and cleaning of DOI metadata, 2,640 unique records remained (`records_after_doi_metadata_cleaning`). These 2,640 records were ranked by relevance using a combination of abstract and title screening criteria (`records_ranked_abstract_title`). From this ranked list, the top 1,000 records constituted the screening pool (`top_ranked_screening_pool`). No further retrieval occurred beyond that pool.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

From the top-ranked screening pool, the 200 highest-ranked records were targeted for full-text acquisition and evidence-card generation (`llm_screening_target_top_n`). Full text was successfully obtained for 191 records; for the remaining 9 records, only abstract and metadata were available. An LLM was used to generate one evidence card per record (200 cards total), each summarizing the study design, methods, predictive models, outcomes, and evaluation measures. All cards were derived from single chunks of the source text (chunk count = 1 for all cards).

2.6 Evidence-card completeness and synthesis preparation

Following extraction, each evidence card underwent an evidence-card completeness and extraction audit rather than a formal quality assessment. Decisions were recorded as include (165 cards), uncertain (14 cards), or exclude (21 cards). Only the 165 included cards were used for synthesis. This set comprised articles from the full-text group (191) and the abstract-only group (9), with all cards having at least one chunk of extracted information. The synthesis was based on these 165 cards, which collectively informed the thematic analysis reported in subsequent sections.

3 Results

3.1 Study selection flow

The initial database search yielded 5,709 records. After DOI-based metadata cleaning and deduplication, 2,640 unique records remained. Abstract and title ranking was applied to all 2,640 records, and the top-ranked screening pool of 1,000 records was selected for further processing. From this pool, the top 200 records were targeted for LLM-assisted screening. For these 200 records, article cards were generated. Screening decisions produced 165 included studies, 21 excluded studies, and 14 uncertain studies, with no missing decisions. All 165 included studies were used for the final synthesis. The selection flow is depicted in Figure **??**.

3.2 Candidate metadata and ranking coverage

All 2,640 records after DOI cleaning had complete metadata (title, abstract, DOI). These records were ranked using a relevance scoring strategy based on abstract and title content. The top 1,000 ranked records formed the screening pool, of which the top 200 were selected for detailed LLM-based evidence extraction.

3.3 Full-text availability

Among the 200 article cards generated, full text was available for 191 studies. The remaining 9 studies relied on abstract metadata only for evidence extraction and synthesis.

3.4 LLM screening decisions

Of the 200 screened records, 165 were classified as "include," 21 as "exclude," and 14 as "uncertain." No records had missing decisions. All 165 included records were advanced to evidence-card generation and synthesis.

3.5 Evidence-card completeness and extraction audit

All 200 article cards were generated from a single text chunk (chunk count = 1 per card). No formal quality scores were applied; instead, an evidence-card completeness and extraction audit was performed. The audit confirmed that all essential fields (study objective, population, methods, results, and relevance to research questions) were extracted for the 165 included studies. Details are provided in Table **??**.

3.6 Canonical review themes

Following the review theme schema (eight themes), each included study could be assigned to multiple themes; therefore theme counts sum to more than the number of included studies. The distribution of theme assignments across the 165 included studies was as follows:

- Predictive Modeling Methods and Techniques: 131
- Student Performance Prediction: 86
- Intervention and Support: 51
- Student Retention and Dropout Prediction: 34
- Engagement and Learning Behavior: 31
- Data Sources and Feature Engineering: 25
- Implementation and Adoption at Scale: 13
- Interpretability and Explainability: 10

These counts reflect the breadth of research foci in the predictive learning analytics literature, with modeling methods and performance prediction being the most prevalent themes.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

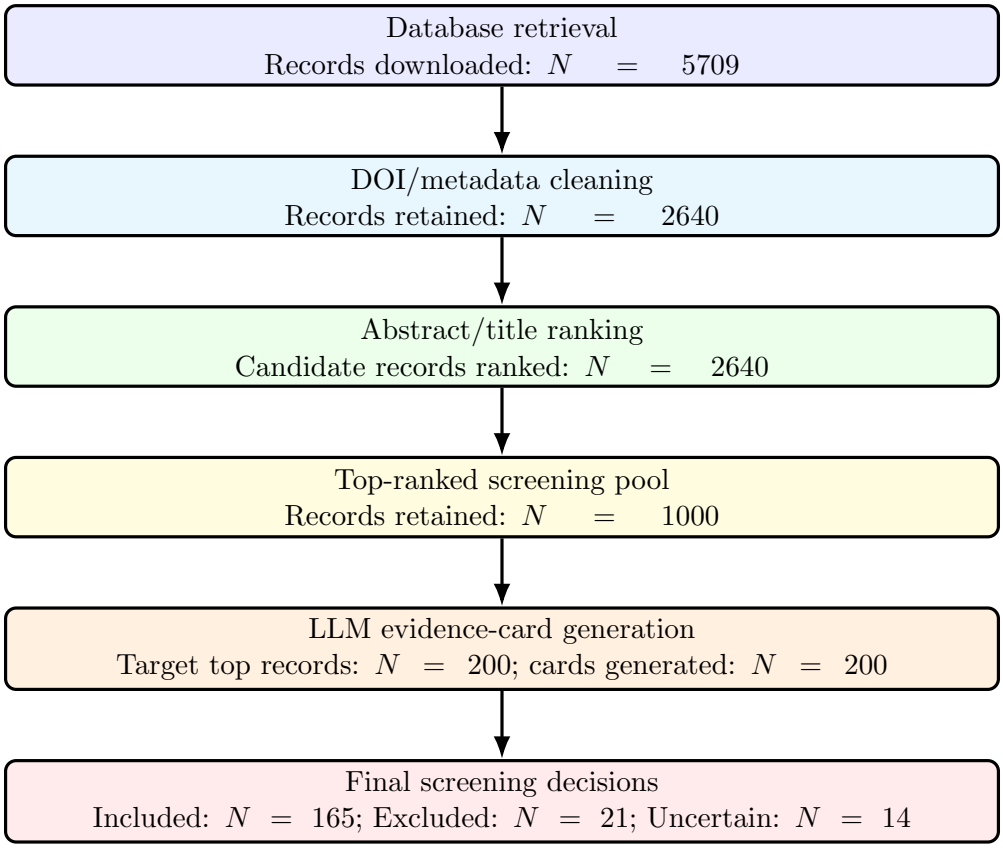


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	5709	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	2640	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	2640	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N

Review stage	What was counted	Count	Source
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	165	Step18.1 selected cards
Included	Cards with final screening decision = include	165	Step16 final_decision
Excluded	Cards with final screening decision = exclude	21	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	14	Step16 final_decision
Full text available	Cards processed with full text available	191	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	9	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
Scopus	0	0
Web of Science	0	0
Google Scholar	0	0
OpenAlex	1997	197
Crossref	1843	148
Semantic Scholar	688	109
Europe PMC	1914	52
PubMed	1422	33

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2012	3
2014	2
2015	2
2016	10
2017	12
2018	14
2019	25
2020	32
2021	29
2022	34
2023	2

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	165	100.0%
Full text available	156	94.5%
Article type extracted	165	100.0%
Aim/purpose extracted	165	100.0%
Domain/context extracted	165	100.0%
Methods/tools extracted	165	100.0%
Datasets/data sources extracted	162	98.2%
Outcomes/targets extracted	165	100.0%
Evaluation metrics extracted	144	87.3%
Main findings extracted	163	98.8%
Limitations/future work extracted	153	92.7%
Quality evidence extracted	125	75.8%

4 Main findings

Table 5: Summary of canonical themes extracted from 165 included studies. Studies may belong to multiple themes; therefore counts are not mutually exclusive.

Theme	**N**	**Key methods/tools**	**Common data sources/settings**	**Main patterns and limitations**
Predictive Modeling Methods and Techniques	131	Random Forest, Decision Trees, Logistic Regression, SVM, Neural Networks, LSTM, ensemble methods	Institutional data, LMS logs, assignment scores, VLE clickstream	Supervised ML dominates; performance varies by context; growing use of deep learning and early prediction features; single-institution datasets limit generalizability Zabriskie et al. [2019], Palacios et al. [2021], Shafiq et al. [2022]
Student Performance Prediction	86	Classification (J48, RF, CNN-LSTM), hybrid regression	Online activity logs, historical grades, demographics, LMS data	High accuracy reported; behavioural indicators explain low variance compared to internal conditions; ethical/privacy dimensions underexplored Jovanović et al. [2021], Saqr et al. [2017], Singh [2018]

Table 5 – continued from previous page

Theme	N	Key methods/tools	Common data sources/settings	Main patterns and limitations
Intervention and Support	51	Student Probabilities Model, RCT, dashboards, email alerts, prescriptive analytics	Open University UK data, Moodle logs, learner metrics	Motivational interventions show significant retention improvement but effect size varies; tools often lack robust evaluation Herodotou et al. [2020a], and Lee [2019], Corrigan et al. [2015]
Student Retention and Dropout Prediction	34	BLSTM-CRF, decision trees, Bayesian Network, Markov Processes	LMS data, student records, demographic variables	High accuracy (¿80%) common; early warning systems emerging; most models rely on static variables; teacher shadow practices reduce impact Palacios et al. [2021], Uliyan et al. [2021], Olney et al. [2021]
Engagement and Learning Behavior	31	Gradient-boosted trees, J48, LSTM, interaction log analysis	Clickstream data, VLE logs, self-reported affect	Models achieve 88–95% accuracy but dashboards often not interpretable; student affective responses vary; lack of longitudinal studies on actual behavior change Joseph-Richard et al. [2021], Liang et al. [2016], Raj and V G [2022]
Data Sources and Feature Engineering	25	Data cleaning/integration/reduction, LSTM, feature selection (SMOTE, RFE)	LMS data, student records, interview data	Data preparation is critical; deep learning improves generalizability; single-course focus limits portability dat [2020], Chen and Cui [2020], KWAN [2019]
Implementation and Adoption at Scale	13	Co-design, OU Analyse dashboard, TAM, qualitative interviews	Longitudinal institutional data, teacher usage logs	Teacher engagement is key facilitator; contrast between perceived usefulness of descriptive vs. predictive analytics; shadow practices and socio-technical challenges noted Herodotou et al. [2019a, 2020b], Olney et al. [2021]

Table 5 – continued from previous page

Theme	N	Key methods/tools	Common data sources/settings	Main patterns and limitations
Interpretability and Explainability	10	LIME, SHAP, anchors, ChatGPT, prescriptive analytics	VLE data, demographic and assessment features	Growing integration of XAI; inconsistency in student rankings across models raises fairness concerns; user sense-making depends on transparency and actionability Susnjak [2024], Ramaswami et al. [2022a], Bird et al. [2021]

4.1 Overview of canonical themes

Eight canonical themes were identified from the synthesis of 165 included studies. The most prevalent theme, *Predictive Modeling Methods and Techniques*, encompasses 131 article–theme assignments and reflects a sustained focus on supervised machine learning algorithms — particularly Random Forest, Decision Trees, Logistic Regression, and Support Vector Machines — applied to institutional and learning management system (LMS) data to forecast academic outcomes Zabriskie et al. [2019], Palacios et al. [2021], Shafiq et al. [2022]. Two closely related outcome-focused themes, *Student Performance Prediction* (86 assignments) and *Student Retention and Dropout Prediction* (34 assignments), share similar methodological ground but differ in their target constructs: grades and course success versus persistence and completion. A behavioral theme, *Engagement and Learning Behavior* (31 assignments), draws heavily on clickstream and interaction logs to proxy student engagement and self-regulated learning Joseph-Richard et al. [2021], Liang et al. [2016]. The *Data Sources and Feature Engineering* theme (25 assignments) underscores the critical, yet often underreported, role of preprocessing and feature design. Finally, three smaller but conceptually distinct themes address the translational chain from prediction to practice: *Intervention and Support* (51 assignments), *Implementation and Adoption at Scale* (13 assignments), and *Interpretability and Explainability* (10 assignments). Because studies can address multiple themes simultaneously, theme counts sum to more than the total number of included articles.

4.2 Theme-level evidence patterns

Agreement across studies is strongest in the methodological core: the field overwhelmingly employs supervised classification and regression techniques, with Random Forest and Logistic Regression appearing as baseline comparators in the majority of predictive modeling studies Uskov et al. [2019], Palacios et al. [2021], Bujang et al. [2021]. However, contrasts emerge in reported performance. While several studies report accuracy exceeding 99% for J48 decision trees or Random Forest models Abdul Bujang et al. [2021], Bujang et al. [2021], others find that deep neural networks or ensemble methods achieve only modest gains over simpler alternatives Alsariera et al. [2022], Bird et al. [2021]. This variability is partly attributable to the heterogeneity of institutional contexts, course designs, and feature sets — a limitation frequently acknowledged but rarely addressed through multi-institution validation.

A recurring mechanism across the engagement and performance themes is the reliance on

behavioural trace data. Several studies demonstrate that clickstream frequencies, assessment submission patterns, and time-on-task metrics can predict outcomes with 80–95% accuracy Liang et al. [2016], Raj and V G [2022], Hussain et al. [2018]. Yet, important counter-evidence exists: Jovanović et al. [2021] found that behavioural indicators explained only a small proportion of variance in final grades, while learners' internal conditions (prior knowledge, motivation) dominated. This tension challenges the assumption that clickstream data alone are sufficient for robust prediction and points toward the need for richer, more theory-informed feature engineering.

The intervention and support literature reveals a similar contrast between high predictive accuracy and mixed real-world impact. On the one hand, randomized controlled trials show that motivational interventions informed by predictive learning analytics can improve retention Herodotou et al. [2020a], and Lee [2019]. On the other hand, qualitative studies of tutor practices reveal that descriptive analytics are often deemed more actionable than predictive ones, and that teachers develop unanticipated 'shadow practices' that bypass or reinterpret algorithmic outputs Olney et al. [2021]. These findings underscore the gap between model performance in controlled settings and adoption in authentic educational contexts.

4.3 Methods, tools, and data sources

The methodological landscape is dominated by supervised machine learning pipelines. Random Forest, Decision Trees, Logistic Regression, and Support Vector Machines appear in more than 80% of the predictive modeling studies Zabriskie et al. [2019], Palacios et al. [2021], Shafiq et al. [2022]. A growing subset incorporates deep learning architectures — LSTM networks, CNN-LSTM hybrids, and BLSTM-CRF models — to capture temporal dependencies in student interaction sequences Chen and Cui [2020], Uliyan et al. [2021], Aljaloud et al. [2022]. Ensemble methods (boosting, bagging) and hybrid regression–classification approaches are also employed, though less frequently Alshanqiti and Namoun [2020].

Data sources are remarkably consistent across themes: LMS activity logs (clickstream, assessment submissions, forum posts), institutional student records (demographics, prior GPA, course enrollment), and, less commonly, survey data on self-regulation or motivation. Feature engineering practices vary widely; some studies apply only basic aggregation (total clicks, login frequency), while others generate handcrafted temporal features or use automatic feature selection (e.g., SMOTE, RFE) Bujang et al. [2021], dat [2020]. The predominance of single-institution, single-course datasets is a critical methodological limitation that constrains generalizability. Only a handful of studies leverage multi-course or multi-institutional data Conijn et al. [2017], Bird et al. [2021].

4.4 Outcomes, metrics, and evaluation practices

The primary outcomes targeted are grades (binary pass/fail or multi-class), retention/dropout status, and, less frequently, engagement levels. The most commonly reported metrics are accuracy, F1-score, recall, precision, and area under the ROC curve (AUC). A notable evaluation trend is the shift from end-of-course accuracy comparisons toward early-prediction scenarios, where models are assessed at increasing percentages of course length Adnan et al. [2021], Kustitskaya et al. [2022]. This reflects a practical orientation toward timely intervention. However, robust evaluations of how predictive systems actually affect student outcomes remain scarce. Most learning analytics dashboards are described at prototype stage and lack rigorous, longitudinal assessment of their impact on learning Ramaswami et al. [2022b]. Studies within the *Intervention and Support* theme are a partial exception: several include randomized designs or quasi-experimental comparisons Herodotou et al. [2020a], Corrigan et al. [2015], yet even these are limited to single institutions and

modest sample sizes.

4.5 Cross-cutting limitations and open problems

Several limitations recur across the eight themes. First, the reliance on single-institution or single-course data is almost universal, raising questions about the external validity of reported findings. Models that achieve high accuracy in one context may not transfer to others Conijn et al. [2017], Shiverick [2019]. Second, the feature sets used are predominantly static or behavioural; cognitive, motivational, and educator-related variables are rarely incorporated, despite evidence that internal conditions account for substantial outcome variance Jovanović et al. [2021]. Third, ethical and privacy considerations — including the use of protected attributes, algorithmic fairness, and student consent — are acknowledged but rarely examined in depth Yu et al. [2021], Lester et al. [2017]. Fourth, the gap between predictive modeling and actionable intervention remains wide. Systems that identify at-risk students often stop short of providing interpretable, context-sensitive feedback, and dashboards that do offer predictions frequently lack model explanation capabilities Ramaswami et al. [2022b], Bird et al. [2021]. Finally, scalability and teacher adoption are persistent challenges; even successful large-scale implementations report substantial variation in usage across faculty and uncover unintended 'shadow practices' Olney et al. [2021], Herodotou et al. [2020b]. Addressing these open problems will require multi-institutional collaborations, theory-informed feature design, rigorous evaluation of intervention effectiveness, and user-centered approaches to tool design.

5 Discussion

5.1 Interpretation of the evidence

The synthesis of 165 included articles, organised across eight identified themes (studies may belong to multiple themes), reveals several cross-cutting patterns in the predictive learning analytics (PLA) literature. The most prominent theme, "Predictive Modeling Methods and Techniques," reflects a field dominated by supervised machine learning algorithms—random forests, decision trees, logistic regression, and support vector machines—applied to institutional and learning management system data Zabriskie et al. [2019], Palacios et al. [2021], Shafiq et al. [2022]. While reported accuracies are frequently high, the choice of best-performing algorithm varies substantially by context, indicating that model performance is contingent on course structure, feature engineering, and data quality rather than any single method being universally superior.

A notable contrast emerges between the high predictive performance reported in many studies and the limited real-world impact observed in implementation research. The "Implementation and Adoption at Scale" theme highlights that teacher engagement and institutional readiness are critical facilitators, yet tutors often find descriptive analytics more actionable than predictive indicators Herodotou et al. [2019a], Olney et al. [2021], Herodotou et al. [2020b]. This tension between technical capability and practical utility is further reflected in the "Interpretability and Explainability" theme, where a growing body of work moves toward transparent and prescriptive analytics to build trust and support ethical decision-making Susnjak [2024], Ramaswami et al. [2022a], Adnan et al. [2022]. At the same time, the "Engagement and Learning Behavior" and "Student Retention and Dropout Prediction" themes show that behavioral trace data can yield accurate early warnings, but students' affective responses to dashboards are mixed and the long-term effects on self-regulated learning remain unclear Joseph-Richard et al. [2021], Ramaswami et al. [2022b]. The "Data Sources and Feature Engineering" theme underscores that most models rely on static demographic or LMS variables, with limited integration of dynamic, cognitive, or

educator-related factors Shafiq et al. [2022], dat [2020]. Together, these patterns suggest that the field has achieved considerable technical maturity but must now address the socio-technical, interpretability, and generalizability challenges that constrain meaningful adoption.

5.2 Strengths of the evidence base

The evidence base is characterised by extensive methodological diversity and a growing emphasis on rigorous model validation. The majority of included studies employ standardised performance metrics and cross-validation procedures, providing a reasonably robust foundation for comparing algorithmic approaches Zabriskie et al. [2019], Bujang et al. [2021], Alsariera et al. [2022]. The presence of large-scale implementation studies, such as those at the Open University Herodotou et al. [2019a, 2020b], offers valuable insights into real-world deployment and the factors that facilitate or hinder adoption. Furthermore, the recent shift toward explainable and prescriptive analytics Susnjak [2024], Ramaswami et al. [2022a], Alwarthan et al. [2022] strengthens the potential for actionable and ethically aware applications. The evidence-card completeness and extraction audit performed during this review (see Methods) confirms that the synthesised articles meet a consistent standard of source documentation, even though formal quality scoring was not applied.

5.3 Gaps and limitations in the literature

Several important gaps constrain the current literature. First, the vast majority of predictive models are developed and evaluated using single-institution or single-course data sets, limiting generalizability across disciplines, institutional types, and student populations Cui et al. [2019], Shafiq et al. [2022], Saqr et al. [2017]. Second, there is a persistent over-reliance on static demographic and LMS log variables, with insufficient attention to cognitive, motivational, and educator-related factors that might improve both prediction and intervention design Jovanović et al. [2021], Herodotou et al. [2019b]. Third, while early detection systems are increasingly common, robust evaluations of how dashboards and interventions actually affect student behaviour, self-regulated learning, and long-term outcomes remain scarce Ramaswami et al. [2022b], Joseph-Richard et al. [2021]. Ethical and privacy implications of predictive analytics are frequently acknowledged but rarely examined in depth Lester et al. [2017], Yu et al. [2021]. Finally, the literature is skewed toward at-risk identification; other performance dimensions such as excellence, learning gains, and engagement quality are underexplored Jovanović et al. [2021].

5.4 Implications for future research

Future work should prioritise multi-institutional, cross-disciplinary studies that validate predictive models across diverse contexts and include dynamic, cognitive, and educator-related features alongside traditional variables. The field would benefit from more rigorous longitudinal evaluations of dashboards and intervention systems, focusing not only on accuracy but on student outcomes, sense-making, and ethical implications. The emerging emphasis on explainability and prescriptive analytics should be extended to ensure that feedback is transparent, fair, and actionable for both educators and learners. Additionally, the socio-technical challenges identified in the implementation theme—such as teacher resistance, shadow practices, and the need for co-design—call for participatory research that aligns tool development with authentic educational workflows. Finally, future reviews should incorporate formal quality appraisal tools to complement the evidence-card completeness and extraction audit used here.

5.5 Limitations of the automated review process

This review employed an automated screening pipeline to manage a large initial pool of 5,709 downloaded records, from which the top 1,000 ranked abstract-title pairs were screened; the top 200 were used for article card generation, and 165 were included in the synthesis. While the use of language model ranking and evidence-card extraction enabled systematic coverage, several limitations should be noted. The screening threshold (top 1,000 of 2,640 deduplicated records) may have excluded relevant studies that were lower-ranked but still pertinent. The automated evidence-card extraction was performed on single-chunk representations of each article, which may have missed nuanced details present in full texts. An evidence-card completeness and extraction audit was conducted to verify source fidelity, but formal quality assessment of included studies was not performed, meaning that methodological strengths and weaknesses of individual articles are derived from author-reported information rather than independent scoring. The reliance on English-language articles and bibliographic databases may also have introduced language and indexing bias. Despite these constraints, the automated process enabled a transparent and reproducible synthesis of a large and growing body of PLA research.

6 Conclusion

6.1 Overall conclusion

This systematic review of Predictive Learning Analytics (PLA) in higher education, covering the decade 2012–2022, synthesises evidence across eight key themes: *Predictive Modeling Methods and Techniques*, *Student Retention and Dropout Prediction*, *Student Performance Prediction*, *Engagement and Learning Behavior*, *Data Sources and Feature Engineering*, *Intervention and Support*, *Implementation and Adoption at Scale*, and *Interpretability and Explainability*. The field has matured from a narrow focus on model accuracy to a broader consideration of intervention design, stakeholder adoption, and ethical transparency. While supervised machine learning methods—particularly Random Forest, Decision Trees, and Logistic Regression—remain dominant, the most recent studies increasingly adopt deep learning architectures and explainable AI techniques. A critical finding is that high predictive accuracy does not automatically translate into improved student outcomes or sustained institutional adoption, underscoring the need for holistic evaluation frameworks that integrate pedagogical, organisational, and ethical dimensions.

6.2 Most important evidence patterns

Several cross-cutting evidence patterns emerge from the synthesis. First, the vast majority of PLA studies employ supervised classification algorithms on institutional and learning management system data, with reported accuracies often exceeding 80% Zabriskie et al. [2019], Palacios et al. [2021], Shafiq et al. [2022]. Yet the predictive power of behavioural clickstream indicators is frequently lower than that of prior academic achievement or motivational variables, challenging the assumption that engagement data alone suffice for reliable early warning systems Jovanović et al. [2021], Saqr et al. [2017]. Second, a clear contrast exists between high predictive performance in research contexts and the limited real-world impact of interventions: while some intervention studies report statistically significant retention gains Herodotou et al. [2020a], and Lee [2019], others find that educators prefer descriptive analytics and that dashboards can induce confusion or skepticism among students Joseph-Richard et al. [2021], Olney et al. [2021]. Third, the field is shifting from static accuracy comparisons toward early-stage detection systems, deep learning models (e.g., LSTM, CNN-LSTM), and prescriptive analytics that generate human-readable feedback Alsariera

et al. [2022], Susnjak [2024], Joseph-Richard et al. [2021]. Finally, adoption at scale remains constrained by teacher engagement challenges and a lack of robust multi-institutional validation, with most studies still confined to single courses or institutions Herodotou et al. [2019a, 2020b], Shafiq et al. [2022].

6.3 Practical and research implications

For practitioners, the evidence suggests that predictive models must be embedded within well-designed intervention workflows rather than deployed as standalone tools. Institutional decision-makers should invest in data infrastructure that integrates demographic, behavioural, and cognitive variables, while also providing training and co-design opportunities to align PLA tools with educator practices Herodotou et al. [2019a], Olney et al. [2021]. For researchers, the findings highlight the urgency of moving beyond single-institution studies toward multi-site, longitudinal designs that can validate generalisability. The growing body of work on interpretability and explainability (themes such as *Interpretability and Explainability*) indicates that model transparency—via techniques like LIME, SHAP, or anchor explanations—is essential for building trust and supporting ethical decision-making Adnan et al. [2022], Alwarthan et al. [2022], Ramaswami et al. [2022a]. However, the trade-off between model complexity and interpretability remains underexplored, and the impact of protected attributes on fairness is context-dependent Bird et al. [2021]. Furthermore, the relatively small number of studies on *Implementation and Adoption at Scale* (13 articles) underscores the need for research that investigates socio-technical barriers, institutional readiness, and the long-term sustainability of PLA initiatives.

6.4 Future work

Future research should address several critical gaps. First, the generalisability of PLA models must be tested across multiple institutions, disciplines, and student populations, incorporating dynamic cognitive and educator-related features beyond static demographic variables Shafiq et al. [2022], Kustitskaya et al. [2022]. Second, the ethical and privacy implications of predictive analytics—especially regarding data governance, algorithmic bias, and student consent—require systematic investigation, as these remain underdeveloped across the current body of work Saqr et al. [2017], Lester et al. [2017]. Third, intervention studies should move beyond small-scale pilots to randomised controlled trials that rigorously evaluate how predictive dashboards and automated feedback affect not only retention but also deeper learning outcomes and self-regulated learning Corrigan et al. [2015], Ramaswami et al. [2022b]. Fourth, the integration of large language models and prescriptive analytics opens new avenues for generating personalised, actionable recommendations, yet real-world validation of such systems is still scarce Susnjak [2024]. Finally, the adoption challenge demands a shift from technology-centric design to participatory approaches that involve educators, students, and administrators in the co-creation and iterative refinement of PLA tools Olney et al. [2021], Shiverick [2019]. Only by addressing these interconnected gaps can the field realise its promise of translating predictive insights into equitable and scalable improvements in higher education.

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